Wed Even

ID No: ______________ Name: ______________________________ Sec No.: ______

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

A

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

22-04-2026 (Wednesday) QUIZ-6 (closed book) Max Marks: 08

Max time: 10 Min

~~Answer only in the box provided~~

Q1. An 8086-based system uses a single 8259A PIC with the following configuration: [2M]

- Base vector address (ICW2) = **60H**

At a given instant, the following conditions exist:

- IRR = **00101010b**
- ISR = **00001000b**
- IMR = **00000000b**

What is the vector number of the interrupt in service ?

63H

— 2M.

Q.2. C1 & C2 of 8254 are configured as event counters (C1 & C2 are independent of each other). Both of them are loaded with an initial value of FFFFH(16-bit counter). Write the instructions to latch the output of both C1& C2 simultaneously. The count of C1 should be available in CX and count of C2 should be available in DX register. Write an ALP to do the needful. Base address of 8254 is A1H(Use base address next possible addresses for 8254 as required) **[2+2+2 M]**

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

Selects Counter		Read/Write Control		Timer Mode		
00	Counter 0	00	Latch Counter	000	Interrupt on T/C	0 – binary 0000, FFFF,
01	Counter 1	01	R/W LSB	001	h/w re-Triggerable one shot	1 – BCD 0000 9999
10	Counter 2	10	R/W MSB	010	rate generator	
11	Read Back Command	11	R/W LSB followed by MSB	011	Square wave generator	
				1x0	s/w triggered strobe	
				1x1	h/w triggered strobe	

A0, A1 = 11 $\overline{CS}$ = 0 $\overline{RD}$ = 1 $\overline{WR}$ = 0

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	1	$\overline{COUNT}$	$\overline{STATUS}$	CNT 2	CNT 1	CNT 0	0

D_5: 0 = Latch count of selected counter(s)
D_4: 0 = Latch status of selected counters(s)
D_3: 1 = Select Counter 2
D_2: 1 = Select Counter 1
D_1: 1 = Select Counter 0
D_0: Reserved for future expansion; Must be 0

```
MOV AL, 110X11X0b      — 1M
out A7H, AL
                         1M
IN AL, A3H
mov cl, al     ] 1M
IN al, A3H
mov ch, al     ] 1M
IN al, A5H
mov dl, al     ] 1M
in al, A5H
mov dh, al.    ] 1M
```

Wed even

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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 2025-26

B

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

22-04-2026 (Wednesday) QUIZ-6 (closed book)

Max time: 10 Min Max Marks: 08

Answer only in the box provided

Q1. An 8086-based system uses a single 8259A PIC with the following configuration: [2M]

- Base vector address (ICW2) = **60H**

At a given instant, the following conditions exist:

- IRR = **00101010b**
- ISR = **00100000b**
- IMR = **00000000b**

What is the vector number of the interrupt in service ?

65_H . — 2M

Q.2. C0 & C2 of 8254 are configured as event counters (C1 & C2 are independent of each other). Both of them are loaded with an initial value of FFFFH(16-bit counter). Write the instructions to latch the output of both C1& C2 simultaneously. The count of C1 should be available in CX and count of C2 should be available in DX register. Write an ALP to do the needful. Base address of 8254 is A1H(Use base address next possible addresses for 8254 as required) **[2+2+2 M]**

D7	D6	D5	D4	D3	D2	D1	D0
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

Selects Counter		Read/Write Control		Timer Mode		BCD
00	Counter 0	00	Latch Counter	000	Interrupt on T/C	0 - binary 0000, FFFF,
01	Counter 1	01	R/W LSB	001	h/w re-Triggerable one shot	1 - BCD 0000 9999
10	Counter 2	10	R/W MSB	010	rate generator	
11	Read Back Command	11	R/W LSB followed by MSB	011	Square wave generator	
				1x0	s/w triggered strobe	
				1x1	h/w triggered strobe	

A0, A1 = 11 $\overline{CS}$ = 0 $\overline{RD}$ = 1 $\overline{WR}$ = 0

D7	D6	D5	D4	D3	D2	D1	D0
1	1	COUNT	STATUS	CNT 2	CNT 1	CNT 0	0

D5: 0 = Latch count of selected counter(s)
D4: 0 = Latch status of selected counters(s)
D3: 1 = Select Counter 2
D2: 1 = Select Counter 1
D1: 1 = Select Counter 0
D0: Reserved for future expansion; Must be 0

```
MOV AL, 110X1X10b   — 1M
Out A7, AL          — 1M

IN AL, A3H
MOV CL, al          ] 1M
IN AL, A3H
Mov ch, al          ] 1M

IN al, A5H
mov dl, al          ] 1M
in al, A5H
mov dl ...          ] 1M
```